

BiDi Coherence-Delta Calculus (CDC): Hybrid State Reduction and Recurrence-Gated Variational Analysis

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Abstract

The BiDi Universal Operator System keeps formal source, native execution, guarded acceptance, recurrence, and evidence on one inspectable path. Within that system, the Coherence-Delta Calculus (CDC) is the formal source language and executable native kernel for guarded hybrid state evolution. Three primitive reductions form its semantic center: synchronous phase flow, balanced-ternary commit, and nested belief/prior exchange. Derived layers provide local observation, durable effects, a guarded Universal Operator (U1), and a variational Universal Operator (U2). U1 accepts a lifted double-cover closure only when reciprocal relation roles, sheet restoration, holonomy, local commits, record/decision agreement, and effect gating all pass. U2 differentiates that exact executed path, then independently verifies complete or explicitly restored recurrence before constructing a monodromy operator or emitting characteristic multipliers. This distinction is operational: the canonical two-turn task loop accepts U1 and yields a source-bound 13-coordinate tangent, but complete native state recurrence is held because latch mode, belief, prior, and unwrapped phase do not all return. Consequently, no canonical-loop multipliers are emitted. A separate calibration fixture with an explicit 4π endpoint restoration and complete derivative yields $M_{\text{rel}} = I_{13}$ and a marginal spectrum. Results bind to canonical source, state manifest, tangent, return and spectrum receipts. Analytic, numerical, negative, sanitizer and permanent-mutant checks test the executable path. Lean and Rocq establish the named finite carrier, event-order, cone, sheet, aperture and barrier statements. Richer continuous semantics and physical interpretations need their own explicit tests.

1 Introduction

Hybrid software often needs continuous evolution, discrete admissibility, nested reference frames, local observation, and effects in one auditable object. BiDi connects those demands through one executable operator path, with explicit contracts between its source, state transitions and downstream analysis. Classical hybrid-automata theory supplies a rigorous setting for continuous modes and discrete transitions [1]. CDC narrows that ambition to a compact executable source contract whose higher-order claims are controlled by receipts.

The contribution is a chain of transformations whose conditions and evidence remain connected:

$$\text{.cdc source} \longrightarrow \text{typed state path} \longrightarrow \mathcal{U}_1 \longrightarrow D\mathcal{U}_1 \longrightarrow \text{return gate} \longrightarrow \text{spectrum}.$$

Each transition may be held when its premises fail, preserving the results already established to its left.

Four contributions follow:

1. a grammar-1 source and native runtime organized around exactly three primitive reductions;
2. U1, a guarded lifted-cover closure that is explicitly weaker than complete-state periodicity;
3. U2, a source-bound tangent and recurrence protocol that distinguishes path derivatives, projected returns, relative returns, monodromy, and Floquet multipliers; and
4. a two-axis evidence discipline separating evidence maturity from execution verdict, backed by deterministic receipts, counterexamples, and finite proof mirrors.

2 Evidence convention

Each claim is reported as

(scope, maturity, verdict, receipt or obligation).

Maturity is one of *specified*, *executed*, *adversarially verified*, or *mechanized finite*. Verdict is one of *accepted*, *held*, *violated*, or *not emitted*. These are independent axes. A recurrence hold may be adversarially verified when the implementation reliably refuses a false periodic-orbit claim. Conversely, a mechanized finite involution says nothing by itself about a continuous physical system.

Execution records identify the tested source and state. Verification checks assess their agreement with the versioned language and ABI contract. The mathematical specification states the broader model and any correspondence still to be established between that model and its implementation.

3 Language and semantic partition

The grammar contains declarations for fields, modules, cells, channels, observers, effects, frameworks, U1, and U2. Its operational core remains:

$$\text{flow}(d), \quad \text{commit}(m), \quad \text{nest}(p, c).$$

Trace/windows, councils, persistence, U1, and U2 are derived jobs.

CDC distinguishes a richer calculus specification from audited native realization v1. The specification may name amplitude relaxation, delayed history, belief flow, plasticity, projected relations, and localized events. The current executor mutates a narrower state. For a selected U1 path, its U2 continuous vector is

$$x = (\theta_1, \dots, \theta_{N_c}, b_1, p_1, \dots, b_{N_m}, p_{N_m}) \in \mathbb{R}^n, \tag{1}$$

ordered by source declaration. The discrete mode is

$$q = (h_1, \sigma_1, \dots, h_{N_c}, \sigma_{N_c}, \text{event itinerary}),$$

where h_i records latch presence and $\sigma_i \in \{-1, 0, +1\}$ is the latch value.

Amplitude, intrinsic frequency, precision, action gain, field constants, and channel parameters are frozen for this tangent calculation. Source topology, jobs, readout, and effect paths are fixed structure. Projected phase, winding, sheet, and declared holonomy are derived values rather than hidden tangent coordinates. The coordinate manifest closes over exactly the executed path; unreferenceed declaration islands cannot add inert unit multipliers.

4 The audited native reductions

4.1 Synchronous phase flow

Native v1 executes one synchronous finite-duration map. For a cell i in the selected field,

$$\theta'_i = \theta_i + \omega_i d + Gd \sum_{e: t(e)=i} w_e \sin(\theta_{s(e)} + \alpha_e - \theta_i), \quad (2)$$

where only same-field channels contribute. Its exact phase Jacobian is

$$\frac{\partial \theta'_i}{\partial \theta_\ell} = \delta_{i\ell} + Gd \sum_{e: t(e)=i} w_e \cos(\theta_{s(e)} + \alpha_e - \theta_i) (\delta_{s(e)\ell} - \delta_{i\ell}). \quad (3)$$

All right sides read the same pre-step state. Beliefs and priors are unchanged. Equation (2) is the implemented map, not an asserted exact solution of an idealized differential equation.

4.2 Balanced-ternary commit

With deadband δ , the current quantizer is

$$\tau(\theta) = \begin{cases} +1, & \cos \theta > \delta, \\ -1, & \cos \theta < -\delta, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The ordered trit walk is accepted only when every prefix sum is nonnegative. An accepted commit writes latches but does not change phase, belief, or prior. In a fixed accepted trit/mode itinerary its continuous reset derivative is therefore identity. A held commit terminates the accepted U1 path. At a quantization boundary, or when a perturbation changes trits or event order, U2 holds rather than differentiating across incompatible modes.

4.3 Nested exchange

Let $u(q)$ be the child mean trit selected from latches when present and current quantization otherwise. Native v1 executes

$$b'_p = b_p + Gu(q), \quad p'_c = b'_p. \quad (5)$$

Within one discrete mode, $u(q)$ is constant. Thus

$$\frac{\partial b'_p}{\partial b_p} = 1, \quad \frac{\partial p'_c}{\partial b_p} = 1, \quad \frac{\partial p'_c}{\partial p_c} = 0.$$

The overwritten child-prior row is not identity. The release includes a finite-difference oracle and a permanent mutant that substitutes an attractive but unexecuted theoretical derivative.

4.4 Specified but not yet executed semantics

The source language can declare delay, amplitude, line projection, precision, and richer guards. Native v1 does not dynamically update amplitude or channel weight, does not carry delay history, does not apply line projection or openness to Equation (2), and does not localize a guard crossing between scheduled operations. These remain specified obligations. In particular, nonzero delay requires an explicit history state, and the current source guard cannot silently authorize saltation.

5 Universal Operator U1

U1 is a derived guarded composition of the three reductions. Let its state component be

$$\mathcal{U}_X : (q_0, x_0) \mapsto (q_T, x_T).$$

An accepted U1 requires reversed-endpoint receptive and radiant channels with a shared pair identity, one-turn projection return with sheet inversion, two-turn projection return with sheet restoration, holonomy agreement, accepted local commits, equality of the computed record and decision coordinates, and enactment of only that accepted coordinate.

This earns *lifted-cover closure*. It does not imply $(q_T, x_T) = (q_0, x_0)$. The distinction is material in the canonical example: the cover projection and sheet close after two turns, while latches populate, context belief and child prior accumulate, and the unwrapped cover phase advances by 4π .

6 Variational Universal Operator U2

6.1 Path tangent

For an accepted path with a stable mode itinerary, U2 may emit

$$D\mathcal{U}_X(x_0) : T_{x_0}X_{q_0} \longrightarrow T_{x_T}X_{q_T}.$$

For column perturbations, operation Jacobians left-multiply in execution order. For smooth segments Φ_k and event maps S_k ,

$$D\mathcal{U}_X = \Phi_{N+1}S_N\Phi_N \cdots S_1\Phi_1. \quad (6)$$

The implementation updates $A \leftarrow J_k A$ from $A = I$. Noncommuting-event fixtures and a reversed-product permanent mutant protect this convention.

6.2 Reset and saltation boundary

Scheduled commits use their reset Jacobian. A state-triggered hybrid event would instead require the sensitivity jump [2, 3]

$$S_e = D_x R_e + \frac{(f^+ - D_x R_e f^- - \partial_t R_e)n^\top}{n^\top f^- + \partial_t g_e}, \quad n^\top = D_x g_e. \quad (7)$$

CDC's generic U2 tests exercise this formula, including transversality and event order. The canonical source path reports its guard method as **unbound-no-saltation**, because its guard is not yet a localized transition surface. This records the current event-localization boundary.

6.3 Recurrence before monodromy

Absolute recurrence requires restored discrete mode, the same coordinate manifest, a compatible itinerary class, and

$$\|x_T - x_0\|_W \leq \varepsilon \quad (8)$$

under recorded absolute and relative tolerances. Lifted coordinates use unwrapped distance; a sheet restoration cannot substitute for the continuous residual.

Relative recurrence requires an explicit executable endpoint restoration $\rho_\gamma : X_{q_T} \rightarrow X_{q_0}$, with

$$\|\rho_\gamma(x_T) - x_0\|_W \leq \varepsilon. \quad (9)$$

The receipt must bind the restoration action, complete derivative, and equivariance/section witnesses. Omitting changed coordinates is only a projected return and cannot authorize Floquet language.

Only a verified return constructs

$$M = D\mathcal{U}_X(x_0), \quad M_{\text{rel}} = D\rho_\gamma(x_T)D\mathcal{U}_X(x_0). \quad (10)$$

Characteristic multipliers are eigenvalues μ_i of the resulting square operator [4]. After independently verified neutral modes are removed, the release classifies a return as stable when all $|\mu_i| < 1 - \tau$, unstable when some $|\mu_i| > 1 + \tau$, and marginal otherwise when at least one mode lies in the unit band. A unit multiplier is not labeled gauge by proximity alone; it requires a declared generator and section alignment.

7 Apertured oriented reciprocity

CDC contains several compatible but distinct oriented structures:

1. carrier inversion exchanges -1 and $+1$ while fixing the aperture 0;
2. receptive and radiant are reversed roles of directed relations;
3. path orientation may reverse angular direction; and
4. one turn flips double-cover sheet parity while two turns restore it.

We call the U-level structure *apertured oriented reciprocity*. It is more precise than a global binary-polarity assertion: reciprocal channels need not cancel, and their residual angular mismatch is retained as holonomy.

The distinctions are necessary. The two-symbol walk $+-$ passes the nonnegative-prefix barrier, while its naive pointwise sign inverse $-+$ fails at the first prefix. Carrier inversion is therefore not a global commit-safety symmetry.

The implemented generic covariance gate accepts a supplied involution only when it fixes the aperture and satisfies both

$$\rho_1\mathcal{U}_X(x) = \mathcal{U}_X^\rho(\rho_0x), \quad D\rho_1D\mathcal{U}_X = D\mathcal{U}_X^\rho D\rho_0. \quad (11)$$

Its analytic fixture and permanent mutants establish the mechanism. No current `.cdc` form binds a conjugate operator to the canonical loop, and no physical polarity law follows.

8 Executable results

Table 1 reports the release’s current decision surface. The machine-readable result is the deterministic line beginning `u2-json=` with schema `cdc.u2.stability.v1`.

The root source registry reports 262 accepted expectations, 20 terms, 22 rules, 16 invariants, 46 capabilities, six framework contracts, and 4831 native witnesses. These are declaration/closure counts, not theorem totals.

Table 1: Receipt-bound U1/U2 results in version 0.3.0.

Scope	Verdict	Evidence and consequence
Canonical U1	accepted	Reciprocal cones, two-turn lifted-cover closure, holonomy, record/decision equality, and guarded enactment pass.
Canonical U2 tangent	accepted	A source-bound, source-ordered 13-coordinate path tangent is emitted.
Canonical full recurrence	held	recurrence-mode-mismatch ; initial and final state/mode remain in the receipt.
Canonical monodromy and multipliers	not emitted	The recurrence gate leaves matrix and spectrum fields null.
Projected cover return	held	undeclared-quotient ; projection is not relative recurrence.
Explicit 4π relative fixture	accepted, marginal	Complete restoration and $D\rho = I$ yield $M_{\text{rel}} = I_{13}$ and thirteen retained physical unit multipliers.
No spectral backend	held	spectral-backend-unavailable ; no eigenvalues are fabricated.

9 Verification and formal layer

The repository-wide gate rebuilds the grammar frontend, native runtime, unified driver, store, and real-Schur path. It checks parser/registry parity, canonical-source identity, replay and generated artifacts, U1 positive and negative cases, U2 analytic and finite-difference oracles, deterministic receipts, effect purity, sanitizers, persistence concurrency, and formal/paper freshness.

The focused U2 gate includes eight permanent mutation families: reversed product order, omitted saltation, omitted endpoint-restoration derivative, invented nest derivative, false recurrence acceptance, ignored aperture, skipped tangent covariance, and false polarity acceptance. Each mutant must be killed by the same tests that accept the unmodified implementation.

The finite Lean and Rocq mirrors establish only their named statements: balanced-ternary counts and finite algebra; tangent identity and associative composition; observable order of two finite noncommuting maps; carrier, cone, and sheet involutions; fixed aperture; and the $+-/-+$ barrier counterexample. They do not prove the runtime Jacobian, recurrence, numerical spectrum, source-bound covariance, continuous holonomy, or a physical theory.

10 Implementation and reproducibility

The path from CDC source to result can be inspected in:

- `kernel.cdc` and `laws.cdc` declare the language contract and 16-key invariant registry;
- `framework_loop.cdc` binds H5, U1, and the canonical U2 chain;
- `runtime/cdc_native_runtime.c` executes primal, U1, persistence, and U2 source jobs;
- `runtime/cdc_variational.c` and `runtime/cdc_linalg.c` implement tangent, recurrence, restoration, polarity, and validated real-Schur logic;
- `tests/u2/` and `tests/fixtures/u2/` contain executable positive, negative, numerical, receipt, and purity cases; and

- `formal/lean/` and `formal/coq/` contain the finite proof mirrors with explicit theorem scopes.

The language/runtime/semantic execution boundary contains one Python file: `cdc_boot.py`, a declaration and expectation oracle. The reducer and U1/U2 analysis execute natively; asset-generation utilities do not execute CDC source. The shipped native ABI is 1.5 and the source grammar is version 1.

11 Limitations and research boundary

Version 0.3.0 does not establish global nonlinear stability, robustness outside the tested neighborhood, arbitrary off-grid event execution, nonzero-delay history dynamics, or parity between prose semantics and unimplemented primal fields. The positive relative fixture validates a return-analysis mechanism; it does not make the canonical loop recurrent.

The implementation is deterministic classical software. Its mathematical state variables acquire a physical interpretation only through a specified mapping to measured quantities, units and constitutive relations. A physical specialization therefore requires calibration data, competing models and discriminating experiments. The software’s recurrence and derivative checks provide tools for that work without establishing the physical mapping itself.

12 Conclusion

CDC connects implemented transitions to the conditions for downstream analysis. U1 binds lifted closure to reciprocal relations, topology, record agreement, and effect gating. U2 binds first-order analysis to the exact executed path and refuses monodromy until state recurrence is independently earned. An accepted cover, an earned tangent and a held recurrence can coexist without being flattened into one verdict. That distinction supports stronger formal and empirical work without assigning it evidence in advance.

Relationship to U and disclosure

The U language develops a broader substrate for operations with explicit laws, resource accounts and observation contracts. CDC predates U and provides an independent compatibility case: its finite phase map, ternary barrier and guarded analysis must retain their existing semantics in any U realization. The projects are available at <https://etellis.github.io/BiDi-web/> and <https://etellis.github.io/U-web/>.

Edward Ellis is the author. AI systems assisted with implementation, mathematical exposition and editorial review; responsibility for the claims remains with the author. This research preprint is prepared for arXiv. No submission or peer review is claimed.

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